AI Solutions Builder

Project Report | Tasks 1–4

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Project overview. A working financial-analysis prototype combining QuickBooks Online Sandbox connectivity through Claude Desktop and MCP, a reusable monthly trend-analysis skill, deterministic Python calculations, automated evaluation and numerical validation, and a documented ten-client architecture.

131	639	9/9	18
Python tests	MCP tests	golden cases	months analyzed

Task 1 | QuickBooks Sandbox + Claude Desktop MCP

Connection, tool invocation and financial-report retrieval

Claude Desktop is connected to a local QuickBooks MCP server configured for the Sandbox environment. Company-information and financial-report tools were used to demonstrate the connection and retrieve a read-only Profit & Loss report.

MCP source	`quickbooks-online-mcp-server/`
Connection notes & evidence	`abc-trend-analysis/docs/ and abc-trend-analysis/results/`
Screenshots	`abc-trend-analysis/docs/screenshots/`

Verification & Evidence (about 2 minutes)

1. Connection evidence: Inspect the screenshots in abc-trend-analysis/docs/screenshots/ showing the Claude Desktop MCP setup and connection status.
2. Workflow evidence: Review the saved artifacts under abc-trend-analysis/docs/ and abc-trend-analysis/results/ that document company-information retrieval and the read-only Profit & Loss workflow.
3. Implementation evidence: Inspect the QuickBooks MCP source, including tool registration and the configured read-only safeguards, to understand how the integration works.
These submitted artifacts allow independent review without access to the author's personal Claude Desktop account. During a live review, the author can share their screen and demonstrate the connection. Reviewers who wish to reproduce the connection independently must configure their own Claude Desktop environment with their own authorized Sandbox credentials. No import or write operation is required for this demonstration.

Task 2 | Reusable Financial Trend Analysis Skill

Deterministic calculation engine, accounting anomaly handling, and CFO review

A reusable Claude skill interprets structured financial statements processed by a deterministic Python calculation engine. The engine applies Decimal cents arithmetic, subtotal isolation, and rolling 12-month TTM gating, while the skill contextualizes variance drivers, isolates the December 2025 non-operating tax credit anomaly (-$1.244M), and formats an actionable CFO Month-End Close Checklist.

Workflow: Excel workbook / saved report → Python engine → results/analysis.json → results/report.md → skill-guided commentary & validation

Reusable skill	`abc-trend-analysis/.claude/skills/monthly-trend-analysis/SKILL.md`
Calculation engine	`abc-trend-analysis/scripts/engine.py`
Source workbook	`abc-trend-analysis/data/ (18-month baseline)`
Generated results	`abc-trend-analysis/results/analysis.json and report.md`

How to verify: Read the skill instructions, inspect scripts/engine.py, trace structured metrics in results/analysis.json to the workbook, and review results/report.md. Run the offline pipeline to reproduce the analysis without changing Sandbox data.

Task 3 | Golden Cases, Regression Tests and Numerical Validation

Repeatable checks for correctness, financial interpretation and grounded output

Nine golden cases cover month ordering, accounting anomalies, TTM eligibility, negative reversals, missing periods, inconsistent totals, and cross-company generalization. Evaluation scripts compare results against expected cases. Commentary checks assess numerical grounding, unsupported causal claims, material anomaly interpretation, and missing-information disclosure. The numerical validator checks claims against generated analysis and labels them VERIFIED, INVALID, or UNVERIFIED.

Golden cases	`abc-trend-analysis/evals/eval_cases.json and evals/fixtures/`
Evaluation scripts	`evals/run_evals.py; evals/eval_commentary.py`
Numerical validation	`evals/numerical_validator.py; evals/llm_inference.py`
Automated tests	`abc-trend-analysis/tests/ (131 automated pytest tests)`

Command (run from abc-trend-analysis/)	Verifies
`pytest`	Python unit, pipeline, adapter, evaluation, and validator tests (131 passed freshly verified).
`python -m evals.run_evals`	Golden-case regression (9/9 cases passed; 110/110 baseline score, 0 regressions).
`python scripts/run_pipeline.py --offline`	Six offline pipeline stages in the recorded run (reconciliation, artifacts, evals).

Verified execution results: 131 Python tests passed; 639 MCP tests passed across 35 suites; 9/9 golden cases passed (110.0/110.0); commentary evaluation 100/100; six offline pipeline stages passed; financial reconciliation 126/126 checks passed (100%).

Task 4 | Architecture and Ten-Client Scaling

A documented design for client isolation and service growth

The architecture note describes a target ten-client design covering tenant identity, authorization, client-scoped credentials and storage, context separation, and operational monitoring. It provides a design path for evolving the prototype into a multi-client service with consistent isolation controls.

Architecture files: abc-trend-analysis/docs/architecture.pdf and abc-trend-analysis/docs/architecture.md. A copy of architecture.pdf is also provided at the submission root.

How to verify: Open the one-page architecture PDF, follow the component/data flow, and read the Markdown companion for tenant identity, authorization, data storage, and credential separation.

Reviewer Walkthrough

A concise, evidence-led order for the demonstration

Stage	Show	Demonstrates
Task 1	MCP status, company-info tool, Profit & Loss report and saved evidence.	Sandbox connection and tool invocation.
Task 2	SKILL.md, engine.py, workbook and generated results.	Reusable workflow and traceable calculations.
Task 3	Test suite, golden-case fixture and numerical validator.	Regression coverage and numerical grounding.
Task 4	Architecture PDF and Markdown companion.	Ten-client design and tenant-isolation approach.

Submission summary. Tasks 1–4 are documented with implementation details, file locations, reviewer verification steps, and recorded test results. The reviewer can inspect the Sandbox workflow, reproduce the financial analysis, rerun evaluations, and review the proposed ten-client architecture.